

High-Resolution ConvNeXtV2 for Identity-Document Fraud Detection

Hanif Noer Rofiq

Kaggle: [hanifnoerr](#)

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Abstract

We describe our submission to the FREUID Challenge 2026, a binary identity-document fraud detection benchmark. The system fine-tunes a ConvNeXtV2-Large classifier at 672×672 resolution using all 69,352 labeled training images. The training recipe is intentionally direct: mild photometric and geometric augmentation, binary cross-entropy, AdamW, cosine learning-rate decay, and mixed-precision optimization. A single model obtained a public leaderboard FREUID score of **0.00267** (lower is better). The private score was unavailable when this report was prepared. We release the training notebook, frozen checkpoint identity, and a network-isolated Docker inference path.

1 Introduction

Identity-document fraud may involve digital editing, generated content, physical manipulation, or recapture through print-and-scan and display-camera pipelines. These processes can alter both semantic content and weak imaging traces. FREUID evaluates a detector on genuine and fraudulent document images, including document domains not represented in the visible training data [1]. For each image x , a method emits a continuous fraud score $p(y = 1 \mid x)$; larger values denote stronger evidence of fraud.

The challenge metric combines the area under the detection-error trade-off curve (AuDET) with APCER at 1% BPCER. The resulting FREUID score lies in $[0, 1]$ and is minimized. This operating regime rewards ranking quality while also penalizing attacks that remain undetected at a low bona-fide error rate.

Our final system uses one high-resolution ConvNeXtV2-Large model. We selected this design because convolutional feature hierarchies provide a useful balance between global document structure and localized texture evidence, while the 672-pixel input retains more manipulation detail than the backbone’s native 384-pixel fine-tuning resolution.

2 Data

We use only the official FREUID training set. It contains 69,352 RGB document images with binary labels: 40,005 bona-fide samples and 29,347 attacks. The visible data cover five country/document-type groups: Egypt, Guinea, Benin, Mozambique, and Mauritius. We do not use external identity-document images, pseudo-labels, or synthetic training samples.

During the public phase, 7,821 test images were available although the sample submission contained 142,818 identifiers. Following the organizer’s protocol, the unavailable private identifiers were assigned placeholders; those values were not evaluated on the public leaderboard. The released Docker pipeline does not use placeholders: it produces one prediction for each image mounted by the organizer.

The only external model resource is the ConvNeXtV2-Large FCMAE checkpoint distributed with `timm`. This initialization was learned with the FCMAE procedure and subsequently fine-tuned on ImageNet classification data [2, 3].

3 Method

3.1 Architecture

ConvNeXtV2 augments the modern ConvNeXt design with Global Response Normalization and is co-designed with fully convolutional masked autoencoding [2]. We instantiate the Large variant through `timm`, replace its classification head with one output logit z , and compute the submitted probability as $\sigma(z)$. No ensembling, auxiliary classifier, metadata feature, threshold calibration, or test-time augmentation is used.

3.2 Image Processing

For training, each RGB image undergoes a bicubic random resized crop to 672×672 with area scale in $[0.88, 1.0]$. Color jitter is applied with probability 0.5 (brightness 0.14, contrast 0.14, saturation 0.05), Gaussian blur with probability 0.12 (kernel 3, $\sigma \in [0.1, 1.0]$), and rotation within $\pm 1.5^\circ$. We then apply ImageNet mean and standard-deviation normalization.

At inference, the shorter image side is resized to 672

Table 1: Frozen training configuration.

Setting	Value
Input resolution	672×672
Epochs	3
Batch size	24
Optimizer	AdamW
Initial learning rate	1×10^{-5}
Weight decay	0.05
Loss	BCE with logits
Schedule	cosine, $T_{\max} = 3$
Gradient clipping	global norm 1.0
Random seed	42
Data-loader workers	12

pixels with bicubic interpolation, preserving aspect ratio, followed by a 672×672 center crop and the same normalization. This deterministic transform is shared by the Colab inference notebook and Docker entry point.

4 Training

The model was trained on one Google Colab G4 runtime exposing 96 GB of GPU memory. The run used all labeled samples (`VAL_TYPE=None`); therefore, no held-out validation metric was used to select the reported checkpoint. This choice maximized the labeled data available to the final public model but makes the leaderboard the only reported generalization measurement.

Table 1 summarizes the frozen configuration. We optimize all network parameters without class weighting. PyTorch automatic mixed precision uses `bfloat16` when supported and `float16` otherwise; gradient scaling is enabled for `float16`. Inputs and model parameters use channels-last memory format, and `TF32` is enabled for supported matrix and convolution operations. The optimizer is AdamW with an initial learning rate of 10^{-5} and weight decay 0.05. A cosine annealing schedule spans the three training epochs. Gradients are clipped to a maximum global norm of 1.0.

Because the final run has no validation partition, the checkpoint named `best.pt` is updated after every epoch. The selected artifact is the epoch-3 checkpoint produced by the frozen notebook.

5 Results

The selected all-train ConvNeXtV2 submission achieved a public leaderboard FREUID score of **0.00267**. Lower scores are better. The private score was unavailable when this report was prepared, so we do not report or infer a private-set result. The submission is produced by a single checkpoint and a single deterministic center-crop inference pass.

6 Reproducibility

The Docker build retrieves the selected checkpoint from public storage and verifies its SHA-256 before embedding it in the image. Runtime inference requires no network access. Dependencies are pinned in `reproducibility/requirements.txt`; the base image is `pytorch/pytorch:2.4.1-cuda12.1-cudnn9-runtime`.

The organizer mounts a flat image directory at `/data`. The container converts each supported image to RGB, applies the deterministic preprocessing, runs sigmoid scoring, and writes exactly two columns, `id,label`, to `/submissions/submission.csv`. The identifier is the input filename stem. Predictions are checked for uniqueness, finiteness, and membership in $[0, 1]$.

```
cd reproducibility
docker build -t freuid-repro .
mkdir -p out
docker run --rm --gpus all \
  --network none \
  -v /path/to/images:/data:ro \
  -v "$PWD/out:/submissions" \
  freuid-repro
```

The reference evaluation target is one NVIDIA A100 40 GB GPU with at most 24 CPU cores. The default Docker batch size is 16 and the worker count is 8; both may be changed through environment variables without changing model outputs.

7 Limitations

The reported score comes from the public split only. The selected run trains on all labels and consequently provides no independent estimate of calibration or out-of-domain performance. It also uses one center crop, which may discard cues near the long image boundary. These choices favor a simple and auditable final pipeline but leave room for document-type holdouts, multi-crop evaluation, and model ensembles in future work.

8 Conclusion

Our submission is a single high-resolution ConvNeXtV2-Large fraud classifier trained on the complete official training set. A small augmentation policy and standard binary optimization were sufficient to obtain a public FREUID score of 0.00267. The accompanying frozen checkpoint, notebook, and network-isolated Docker image specify the complete path from an input image directory to the submission file.

9 Artifact Manifest

The public implementation is available on GitHub. Model and training code were frozen before the private

test release. The identifiers below define the exact reported system.

Code freeze

freuid-code-freeze-2026-07-13
Commit 707309a0fb08b176858bfb4e9909b45ef728cdb5

Model identifier

convnextv2_large.fcmae_ft_in22k_in1k_384

Training notebook

notebooks/freuid_convnextv2_g4_672_alltrain_best_public.ipynb
SHA-256: eeeecd050a11e06bfd4ede9f930c4807564867c0c30ec7432029c5939e20fa57

Selected checkpoint

Epoch 3, file `best.pt`, approximately 2.2 GB.
SHA-256: bb3870c09ec7fa21255df1cff40b6cd2ff5a7290904562d8e43a284e4ba5c41a

Checkpoint storage

<https://drive.google.com/drive/folders/15pj--qzgWVzq-KjH7kJUcxnTS3ZA5c2P>

Reported submission

submission_convnextv2_g4_672_alltrain.csv

Container contract

Input: flat image directory at `/data`.

Output: `/submissions/submission.csv`.

Columns: `id,label`; one finite score per input image.

References

- [1] I. Relić, V. D’Elia, S. Bortoli, R. Tudoran, H. Nedyalkova, M. Bošnjak, S. Pavlić, T. Mavračić, D. Dašić, M. Kačan, J. Šegvić, P. Čerić, F. Šoprun, M. Miočić, M. Oršić, L. Vaiani, and P. Garza, “The FREUID Challenge 2026 - IJCAI-ECAI.” <https://kaggle.com/competitions/the-freuid-challenge-2026-ijcai-ecai>, 2026. Kaggle.
- [2] S. Woo, S. Debnath, R. Hu, X. Chen, Z. Liu, I. S. Kweon, and S. Xie, “ConvNeXt V2: Co-designing and scaling convnets with masked autoencoders,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16133–16142, 2023.
- [3] R. Wightman, “PyTorch Image Models.” <https://github.com/huggingface/pytorch-image-models>, 2019.